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Chao Jin, Wen Long, Linna Wang, Zhikang Liu & Liangliang Sun

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physical exercise on the mobile phone addiction and academic
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Chao Jin^{†*} Wen Long[†] Linna Wang Zhikang Liu Liangliang Sun

Chao Jin^{†*}

Office of Academic Research, Jining University, Qufu City, Shandong
Province,

Zip: 273155, China

Email: jnxyjc@126.com

ORCID: <https://orcid.org/0000-0002-9628-9480>

Wen Long[†]

College of Teacher Education, Jining University, Qufu City, Shandong
Province,

Zip: 273155, China

Email: 3141229258@qq.com

ORCID: <https://orcid.org/0009-0008-8263-7828>

Linna Wang

Nishan Middle School, Qufu City, Shandong Province, Zip: 273155, China

Email: nszxlinda@126.com

ORCID: <https://orcid.org/0009-0000-4375-9082>

Zhikang Liu

College of Biological and Chemical Engineering, Qilu Institute of
Technology, Jinan

Zip: 250200, China

Email: 1379628439@qq.com

ORCID: <https://orcid.org/0009-0005-0569-5109>

Liangliang Sun

College of Basic Medical Sciences, Jining Medical University, Jining City,
Shandong Province, Zip: 272067, China

Email: 360212109@qq.com

ORCID: <https://orcid.org/0009-0002-0445-773X>

Note: (1) †: These authors contributed equally to this work and shared first authorship. *: Chao Jin is the corresponding author.

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The mediating and moderating effects of learning engagement and physical exercise on the mobile phone addiction and academic burnout

Abstract

This study was designed to explore the relationship between mobile phone addiction and academic burnout among Chinese college students. We employed the Mobile Phone Dependence Index Scale (MPDIS), Utrecht Work Engagement Scale-Student (UWESS), Maslach Burnout Inventory-Student Survey (MBISS), and Physical Exercise Rating Scale (PARS-3) to conduct a questionnaire-based survey among 700 Chinese college students. Ultimately, 677 valid questionnaires were obtained. There was a significant positive correlation between Chinese college students' mobile phone addiction and academic burnout ($r = 0.40, p < 0.01$), and a significant negative correlation between mobile phone addiction and learning engagement ($r = -0.18, p < 0.01$). Additionally, learning engagement was significantly negatively correlated with academic burnout ($r = -0.57, p < 0.01$). Learning engagement played a partial mediating role between mobile phone addiction and academic burnout, accounting for 20.7% of the total effect. Physical exercise also moderated the relationship between mobile phone addiction and learning engagement. These findings contribute to a better understanding of the mechanisms underlying the relationship between mobile phone addiction and academic

burnout. The discussion provides practical and effective recommendations for the prevention and intervention for academic burnout among Chinese college students. However, the present study is limited by its cross-sectional design and single-province sample, and future research should adopt longitudinal or multi-regional designs to further validate and extend the proposed model.

Keywords: Mobile phone addiction; Learning engagement; Academic burnout; Physical exercise; Undergraduates

Introduction

Academic burnout refers to a persistent, negative, learning-related psychological state characterized by emotional exhaustion, cynicism, and reduced efficacy, which commonly occurs among students¹⁻⁵. Recent meta-analyses have confirmed that academic burnout significantly impairs learning motivation, learning engagement, and psychological well-being, thereby influencing the overall quality of higher education. In the digital era, higher education plays a crucial role in national competitiveness^{6,7}. Promoting students' academic success and mental health has thus become a key component in cultivating innovative, high-quality talent and sustaining educational excellence^{8,9}. Therefore, understanding the factors that contribute to academic burnout among college students is of both theoretical and practical importance^{10,11}.

In addition, based on the Interaction of Person-Affect-Cognition-Execution model (I-PACE) and the Conservation of Resources Theory (COR), mobile phone addiction depletes attentional and self-regulatory resources, resulting in resource loss that heightens fatigue and burnout^{12,13}. Within this framework, decreased learning engagement functions as a mediating mechanism linking excessive mobile phone use to academic burnout. Recent national data show that, by December 2022, the number of Internet users in China reached 1.067 billion, of whom 99.8% accessed the Internet via mobile phone¹⁴. In this context, some college students fall into patterns of compulsive phone use that interfere with academic performance and well-being. Students who exhibit higher levels of mobile phone addiction are often distracted, display diminished learning motivation, and show lower academic efficiency, which collectively heighten their risk of experiencing academic burnout¹⁵. However, this relationship is not direct or singular; rather, it is likely to be mediated by students' engagement in learning activities.

Learning engagement refers to students' vigorous, dedicated, and absorbed participation in academic activities¹⁶, representing behavioral, emotional, and cognitive investment in learning^{17,18}. As an essential indicator of learning quality, learning engagement reflects the degree to which students invest effort and emotion in academic activities. Empirical evidence suggests that excessive mobile phone use reduces time and cognitive resources for study,

leading to a decline in engagement and, consequently, greater burnout¹⁹. When learning engagement declines, students are more likely to experience fatigue, inefficacy, and disengagement, forming a pathway from mobile phone addiction to academic burnout.

Physical exercise is planned, structured, and repetitive physical activities aimed at improving or maintaining physical fitness and psychological well-being²⁰. Recent evidence indicates that regular physical exercise enhances sleep quality, cognitive performance, and executive functioning among adults^{21,22}. Such improvements contribute to better academic readiness and emotion regulation, thereby buffering the adverse impact of mobile phone addiction on learning engagement.

In addition, physical exercise is an important activity in Chinese college students' daily lives. Nonetheless, whether and how physical exercise moderates the relationship between mobile phone addiction and learning engagement remains insufficiently studied. On the one hand, engaging in regular exercise may improve students' concentration and self-control, reducing the disruptive effects of mobile phones. On the other hand, the positive emotions generated by physical exercise may enhance learning interest and motivation, thereby improving engagement levels.

Although previous research has examined mobile phone addiction, learning engagement, physical exercise, and academic burnout independently, the integrative mechanism linking these variables remains unclear. This study therefore constructs a

moderated mediation model to investigate (a) the mediating effect of learning engagement between mobile phone addiction and academic burnout, and (b) the moderating role of physical exercise in this relationship. By clarifying these mechanisms, this research provides theoretical insight and empirical evidence for developing interventions aimed at reducing burnout and promoting students' physical and mental health in higher education.

Literature Review

Relationship between mobile phone addiction and academic burnout

In the digital age, mobile phone addiction has become a common and concerning addictive behavior among college students²³⁻²⁵. A substantial number of studies have indicated that mobile phone addiction has a significant direct impact on academic burnout²⁶⁻²⁸. From the perspective of learning efficiency and performance, mobile phone addicts often check social media frequently or receive irrelevant notifications and information. This behavior seriously distracts attention, making it difficult to concentrate while studying, greatly reduces learning efficiency, and significantly negatively affects learning performance²⁷. According to Cognitive Load Theory, continuous distraction of attention leads to cognitive overload²⁹. Studies have shown that college students' working memory capacity decreases by about 30% after using mobile phones for more than 5 hours per day, which directly weakens their ability to complete complex learning

tasks²⁷. In addition, frequent use of mobile phones may increase students' social pressure and cause anxiety, which will further reduce their learning motivation and worsen their learning status. Longitudinal tracking data show that the anxiety level of high mobile phone addiction groups increased by 42% compared with the baseline in the middle and late semester, which significantly reduced their learning motivation³⁰.

From the perspective of sleep and mental state, using mobile phones for a long time at night will interfere with students' sleep patterns. Sleep is an important basis for restoring energy and maintaining a good learning state. Once the quality of sleep declines, students' mental state will be seriously affected the next day, and their fatigue will be significantly enhanced. In this state of physical and mental exhaustion, students are more likely to have burnout in the learning process, which leads to academic burnout^{31,32}. Therefore, this paper puts forward Hypothesis 1: mobile phone addiction positively predicts academic burnout.

The Mediating Role of learning engagement

There is a significant correlation between mobile phone addiction and learning engagement³³⁻³⁵. As a key construct in positive psychology, learning engagement comprises vigor, dedication, and absorption¹. Previous studies have shown that excessive use of mobile phones for extended periods seriously distracts students' attention, making it difficult for them to

concentrate on learning tasks and fully devote themselves to learning, which leads to a decrease in learning engagement^{30,36}.

Moreover, learning engagement and academic burnout are two closely related variables. When students' learning engagement is high, they can participate more actively in the learning process, which is helpful in effectively reducing learning pressure and anxiety and improving self-efficacy³⁷. The improvement of self-efficacy will further promote students' positive attitude toward learning, make them more confident and motivated when facing learning tasks, and then show a lower degree of academic burnout^{38,39}. Therefore, Hypothesis 2 is put forward: Learning engagement plays an intermediary role in predicting academic burnout by mobile phone addiction.

The Moderating Role of physical exercise

Physical exercise is a physical activity aimed at fitness, entertainment, and health care, strengthening individual physiques and improving health⁴⁰. Physical exercise not only improves mood and reduces anxiety but also enhances psychological resilience and emotional stability through improving self-regulation and neuroplasticity⁴¹⁻⁴³. Furthermore, exercise enhances attention control, executive functioning, and neurocognitive plasticity, helping students focus and manage distractions⁴⁴⁻⁴⁶. Within the COR framework, physical exercise restores depleted resources and supports self-regulation, enabling students to counteract the cognitive depletion caused by excessive phone use. When students

develop regular exercise habits, they can better inhibit distraction, allocate cognitive resources efficiently, and sustain engagement. However, few studies have examined the moderating role of physical exercise within the cognitive-motivational process that leads to academic burnout. In particular, prior mediation and moderation models seldom incorporate physical exercise as a protective factor that replenishes psychological resources and strengthens self-regulation. Therefore, Hypothesis 3 is proposed: physical exercise moderates the relationship between mobile phone addiction and learning engagement. The construction process of the moderated mediation model is shown in Figure 1.

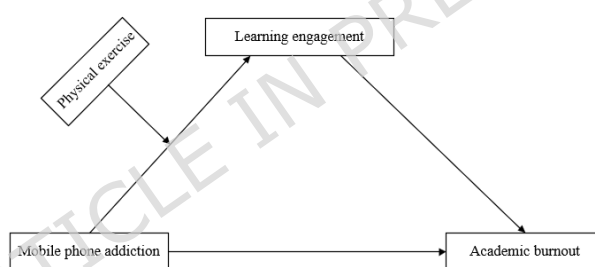


Figure 1. Hypothetical moderated mediation model.

Materials and methods

Participants

In this study, a total of 700 Chinese college students were selected from five universities in Shandong Province using a cluster sampling method, and this survey was administered in March 2025. All five participating institutions were ordinary universities, including three comprehensive universities and two normal universities, with disciplines and majors distributed across arts and

humanities, social sciences, natural sciences, and engineering. During the survey, the group test was administered in classroom settings. Prior to the survey, researchers read the informed consent form to participants, emphasizing the principles of voluntariness, anonymity, and truthfulness, to ensure that they fully understood the study's purpose, methods, potential risks, and benefits. After completing the questionnaire, participants returned it on the spot. Participants could stop answering at any time for any reason, and all data were processed anonymously, with no identifying information. This research was approved by the Ethics Committee of Jining University (2025JNXYLL-017), and all research was performed in accordance with relevant guidelines. A total of 700 paper questionnaires were distributed. After screening, 23 questionnaires with short answer time and extreme values were eliminated, resulting in 677 valid questionnaires for analysis, with an effective rate of 96.7%. Among the respondents, there were 311 boys (45.9%) and 366 girls (54.1%). The sample included 269 freshmen (39.7%), 179 sophomores (26.4%), and 229 juniors (33.8%).

Methods

Mobile Phone Dependence Index Scale (MPDIS)

Compiled by Leung ⁴⁷ at the Chinese University of Hong Kong, the scale was revised for the Chinese context by Huang et al.⁴⁸. The Mobile Phone Dependence Index Scale (MPDIS) consists of 17 items and is divided into four dimensions: loss of control,

withdrawal, avoidance, and inefficiency. Loss of control refers to the desire and behavior that individuals find it difficult to control their own use and collection, corresponding to items 1–7, such as “Your friends and family have complained because you are using mobile phones”; Withdrawal refers to a series of discomfort symptoms such as anxiety, irritability and uneasiness when individuals reduce or stop using mobile phones, corresponding to items 8–11, such as “If there is no mobile phone signal, you will feel irritable”; Avoidance refers to individuals using mobile phones to escape from stress, negative emotions or unpleasant scenes in real life, corresponding to items 12–14, such as “When you feel isolated, you will chat with others with your mobile phone”; Inefficiency refers to the negative impact of excessive use of mobile phones on the efficiency of individual daily activities, corresponding to items 15–17, such as “You find yourself addicted to mobile phones when you have other things to do, which brings you some trouble.” The scale uses the Likert 5-point scoring method, and the higher the score, the greater the dependence on mobile phones. The Cronbach’s α coefficient of the scale is 0.87.

Utrecht Work Engagement Scale-Student (UWES-S)

Adapted from Schaufeli et al.¹, the scale was revised for the Chinese context by Fang et al.⁴⁹. The scale contains 17 items divided into three dimensions: vigor, dedication, and absorption. The vigor dimension reflects strong learning motivation and perseverance, corresponding to items 1–6, such as “I can continue

to study for a long time without taking breaks”; The dedication dimension indicates that individuals recognize the value of learning and maintain enthusiasm, corresponding to items 7–11, such as “I am proud of my learning achievements”; The absorption dimension refers to the state of concentration and immersion in learning, corresponding to items 12–17, such as “I feel very happy when I am fully engaged in studying.” The scale uses a Likert 5-point scoring method, with higher scores indicating higher levels of learning engagement. The Cronbach’s α coefficient of this scale is 0.96.

Maslach Burnout Inventory-Student Survey (MBISS)

Adapted from Schaufeli et al.¹, the scale was revised for the Chinese context by Wu et al.⁵⁰. The scale consists of 16 items divided into three dimensions: emotional exhaustion, cynicism, and academic efficacy. The exhaustion dimension refers to the fatigue and depletion caused by learning, including four items: 2, 5, 8, and 12, such as “I feel empty recently and don’t know what to do”; The cynicism dimension reflects a negative attitude towards learning, including five items: 3, 6, 9, 10 and 13, such as “I don’t think I understand anyway, and it doesn’t matter whether I learn or not”; The inefficacy dimension assesses the individual’s low sense of accomplishment in learning, using reverse scoring for 7 items: 1, 4, 7, 11, 14, 15 and 16, such as “I can always cope with learning problems easily.” The scale uses a Likert 5-point scoring method, with higher scores indicating more severe academic burnout. The Cronbach’s α coefficient of the scale is 0.78.

Physical Exercise Rating Scale (PARS-3)

The Physical Activity Rating Scale (PARS-3), originally compiled by Hashimoto et al.⁵¹ and revised by Liang et al.⁵², was used to assess participants' physical activity levels. This scale employs a 5-point scoring system to evaluate exercise volume based on three dimensions: exercise intensity, exercise duration, and exercise frequency. Each dimension is scored from 1 to 5 points, with higher scores indicating greater exercise volume. Exercise volume is calculated using the formula: Exercise Volume = Exercise Intensity $\times$ (Exercise Duration - 1) $\times$ Exercise Frequency, with a score ranging from 0 to 100 points. The levels of exercise volume are categorized as follows: low exercise volume (≤ 19 points); moderate exercise volume (from 20 to 42 points); and high exercise volume (≥ 43 points).

Statistical Processing

All statistical analyses were performed using SPSS 25.0 software. First, we tested for method bias to check for potential biases caused by using self-report questionnaires. Second, we performed descriptive statistics and correlation analyses, including means, standard deviations, and correlation coefficients among all variables. Finally, to verify our hypotheses, we used the PROCESS macro plugin (Model 7) in SPSS to analyze the relationships between variables⁵³. The PROCESS macro plugin was based on 5,000 bootstrap resampling evaluations of model testing and 95% confidence interval (95% CI) estimates, with a relationship

considered significant when the 95% CI did not include 0. Gender and age were controlled as covariates in the analysis. The significance level was set at $\alpha = 0.05$ ⁵⁴. Additional analyses and detailed measurement information are provided in the Supplementary Data.

Results

Common Method Deviation Test

To assess potential common method bias (CMB), the Harman single-factor test was conducted following Podsakoff et al.⁵⁵. Common method bias refers to the artificial covariance between predictor and criterion variables due to shared data sources, raters, measurement environments, and item characteristics⁵⁶. In this study, questionnaires were designed with anonymous completion and reverse-scored items to mitigate CMB. The Harman single-factor test revealed 11 factors with eigenvalues greater than 1. The first factor accounted for 23.93% of the variance, which is below the critical threshold of 40%. These results indicate that common method bias is not a significant concern in this study⁵⁷.

Descriptive Statistics and Correlation Analysis

Table 1 displays the descriptive statistics for all study variables. For mobile phone addiction, female participants had higher mean scores than male participants. Freshmen had the highest mean scores, while juniors had the lowest. Participants with below-average academic performance had the highest mean scores, whereas those with above-average performance had the lowest. For

learning engagement, female participants had higher mean scores than male participants. Juniors had the highest mean scores, while freshmen had the lowest. Participants with above-average academic performance had the highest mean scores, while those with below-average performance had the lowest. For academic burnout, male participants had higher mean scores than female participants. Freshmen had the highest mean scores, while juniors had the lowest. Participants with below-average academic performance had the highest mean scores, whereas those with above-average performance had the lowest. For physical exercise, male participants had higher mean scores than female participants. Sophomores had the highest mean scores, while juniors had the lowest. Participants with medium academic performance had the highest mean scores, while those with above-medium performance had the lowest.

Table 1. *Descriptive statistics and corresponding differences across demographic characteristics for latent variables*

Variable	MPA	LE	AB	PE
Total	47.41(9.06)	79.11(14.35)	43.22(6.73)	14.13(13.60)
Gender				
Male	47.30(8.83)	76.33(14.79)	44.50(6.27)	17.18(15.61)
Female	47.49(9.26)	81.48(13.54)	42.13(6.92)	11.53(11.00)
Grade				
Freshman	48.87(9.25)	77.24(13.93)	44.06(6.53)	15.47(13.80)
Sophomore	47.94(9.33)	78.05(14.25)	43.96(6.87)	15.87(14.66)
Junior	45.27(8.20)	82.14(14.48)	41.66(6.59)	11.19(11.98)
Academic Performance				
Above-average	47.25(8.83)	82.36(14.32)	40.98(6.49)	13.32(12.91)
Intermediate	47.35(9.04)	78.63(13.93)	43.28(6.49)	14.60(13.50)
Below-	47.82(9.50)	75.40(14.61)	46.56(6.41)	14.05(14.95)

average

Note: MPA = Mobile Phone Addiction; LE = Learning Engagement; AB = Academic Burnout; PE = Physical Exercise

Table 2. *Descriptive statistics and correlations among variables (N = 677)*

Variables	M $\pm$ SD	MPA	LE	AB	PE
MPA	47.41 $\pm$ 9.06	1			
LE	79.11 $\pm$ 14.35	0.18**	1		
AB	43.22 $\pm$ 6.73	0.40**	0.57**	1	
PE	14.13 $\pm$ 13.60	-0.05	0.25	-0.06	1

Note: MPA = Mobile Phone Addiction; LE = Learning Engagement; AB = Academic Burnout; PE = Physical Exercise. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Descriptive statistics and correlation analysis among variables are shown in Table 2. The results indicate that there is a significant positive correlation between mobile phone addiction and academic burnout ($r = 0.40$, $p < 0.01$), and a significant negative correlation between mobile phone addiction and learning engagement ($r = -0.18$, $p < 0.01$). Additionally, learning engagement is significantly negatively correlated with academic burnout ($r = -0.57$, $p < 0.01$). There is no significant correlation between physical exercise and mobile phone addiction, learning engagement, and academic burnout.

Mediating Effect of learning engagement between mobile phone addiction and academic burnout

Table 3. *Mediation model tests*

Variable	Model 1: AB	Model 2: LE	Model 3: AB
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	β	SE	t	β	SE	t	β	SE	t
Gender	-1.21	0.40	-2.99**	5.16	1.08	4.77***	-1.21	0.40	-2.99**
Grade	-0.16	0.24	-0.66	1.65	0.63	2.60**	-0.16	0.24	-0.66
MPA	0.23	0.02	10.12***	-0.24	0.06	-4.11***	0.23	0.02	10.12***
LE							-0.23	0.01	-16.03***
R²		0.16			0.10			0.64	
F		127.65***			14.47***			120.91***	

Note: MPA = Mobile Phone Addiction; LE = Learning Engagement; AB = Academic Burnout; PE = Physical Exercise; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The mediating effect of learning engagement was tested using Model 4 of the PROCESS macro in SPSS 25.0. In all mediation and moderated mediation analyses, *gender* and *age* were entered as covariates to control for demographic influences. The inclusion of these variables did not change the direction or statistical significance of the main effects. Results shown in Table 3 indicate that mobile phone addiction can directly and significantly predict academic burnout ($\beta = 0.23$, $SE = 0.02$, $p < 0.001$). Mobile phone addiction had a significant negative effect on learning engagement ($\beta = -0.24$, $SE = 0.06$, $p < 0.001$), and learning engagement had a significant negative effect on academic burnout ($\beta = -0.23$, $SE = 0.01$, $p < 0.001$). Additionally, the Bootstrap test results showed that the mediating effect value was 0.06, with a 95% confidence interval of [0.03, 0.09]. Since the confidence interval did not include zero, the mediating effect was considered significant⁵⁸. Thus, mobile phone addiction not only directly predicts academic burnout but also indirectly predicts academic burnout through the

mediating effect of learning engagement. The direct effect (0.23) and mediating effect (0.06) accounted for 79.3% and 20.7% of the total effect (0.29), respectively.

When both mobile phone addiction and learning engagement entered the regression equation, mobile phone addiction significantly negatively predicted learning engagement ($\beta = -0.24$, $SE = 0.06$, $p < 0.001$); learning engagement could significantly negatively predict academic burnout ($\beta = -0.23$, $SE = 0.01$, $p < 0.001$). The bias-corrected percentile Bootstrap test showed that the mediating effect between mobile phone addiction and academic burnout was significant, $a * b = 0.06$, $SE = 0.02$, and the 95% confidence interval was [0.03, 0.09]. The mediating effect accounted for 20.7% of the total effect.

Analysis of the Moderating Effect of physical exercise between mobile phone addiction and academic burnout

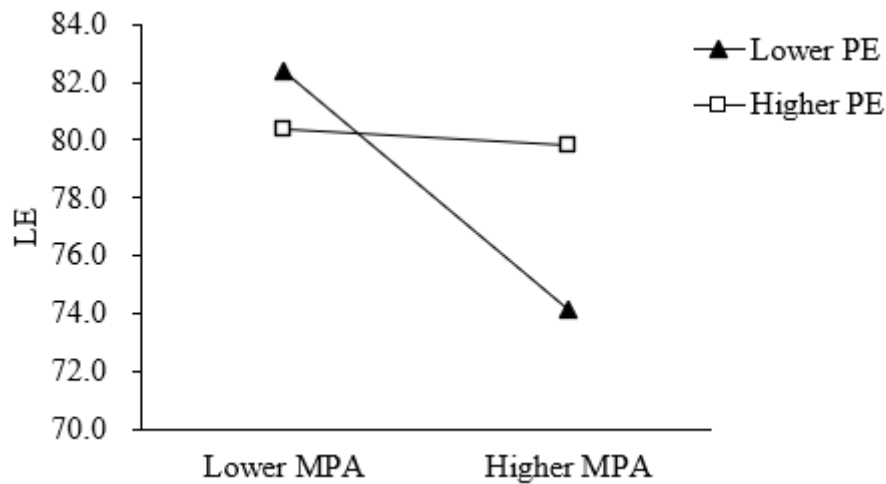
The moderating effect of physical exercise was tested using Model 7 of the PROCESS macro in SPSS. As shown in Table 4, the interaction term between physical exercise and mobile phone addiction significantly predicted learning engagement ($\beta = 0.16$, $SE = 0.01$, $p < 0.001$), indicating that physical exercise moderates the relationship between mobile phone addiction and learning engagement; however, the interaction term did not significantly predict academic burnout ($\beta = 0.00$, $SE = 0.00$, $p > 0.05$), suggesting that physical exercise does not moderate the relationship between mobile phone addiction and academic burnout. To further

explore the interactive effect between mobile phone addiction and physical exercise, participants were divided into high and low physical exercise groups based on the mean $\pm$ standard deviation. A simple slope test was conducted, and the results are illustrated in Figure 2. The results showed that in the high physical exercise group, mobile phone addiction did not significantly predict learning engagement ($\beta = -0.03$, $SE = 0.09$, $p > 0.05$); In contrast, in the low physical exercise group, mobile phone addiction had a significant negative effect on learning engagement ($\beta = -0.46$, $SE = 0.08$, $p < 0.001$). These findings indicate that physical exercise significantly moderates the relationship between mobile phone addiction and learning engagement, with higher levels of physical exercise alleviating the negative impact of mobile phone addiction on learning engagement.

Table 4. *Moderated mediation model tests*

Variable	Model 1: LE				Model 2: AB			
	β	SE	t	95% CI	β	SE	t	95% CI
Gender	5.16	1.08	4.77***	[3.04, 7.29]	-1.21	0.40	-2.99**	[-2.00, -0.41]
Grade	1.65	0.63	2.60**	[0.41, 2.89]	-0.16	0.24	-0.66	[-0.62, 0.31]
MPA	-0.27	0.06	-4.49***	[-0.39, -0.15]	0.23	0.02	54.08***	[0.18, 0.27]
PE	0.01	0.04	0.35	[-0.06, 0.09]	0.02	0.02	-1.27	[-0.05, 0.01]
MPA * PE	0.16	0.01	3.54***	[0.01, 0.03]	0.00	0.00	-0.05	[0.00, 0.00]
LE					-0.23	0.01	-1.71***	[-0.27, -0.21]
R²			0.06				0.41	
F			13.13***				117.49***	

Note: MPA = Mobile Phone Addiction; LE = Learning Engagement; AB = Academic Burnout; PE = Physical Exercise; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.



Note: MPA = Mobile Phone Addiction; LE = Learning Engagement; AB = Academic Burnout; PE = Physical Exercise

Figure 2. The simple slope test illustrates the moderating effect of PE on the relationship between MPA and LE.

Discussion

This study explored the structural relationship between Chinese college students' mobile phone addiction and academic burnout, and verified the mediating role of learning engagement and the moderating role of physical exercise between mobile phone addiction and learning engagement. From a theoretical standpoint, the findings were interpreted within an integrated framework that combines the I-PACE model and the COR, illustrating how excessive mobile-phone use depletes attentional and self-regulatory resources and thereby increases the risk of burnout through reduced engagement.

The Direct Impact of mobile phone addiction on academic burnout

The results of this study indicate a significant positive correlation between mobile phone addiction and academic burnout,

suggesting that higher levels of mobile phone addiction are associated with more severe academic burnout. These findings are consistent with Hypothesis 1 and align with previous research²⁶⁻²⁸. Furthermore, these findings further support the resource-depletion perspective proposed by Cognitive Load Theory and COR Theory. Cognitive Load Theory emphasizes that excessive mobile-phone use consumes limited attentional and working-memory resources, thereby reducing students' cognitive capacity for academic tasks. In contrast, the COR Theory focuses on the depletion of broader psychological resources, such as motivation, emotional energy, and self-regulatory capacity. Integrating these two perspectives illustrates how cognitive overload and psychological resource loss jointly contribute to academic burnout, enhancing the explanatory power of the present model. Notably, different subdimensions of mobile phone addiction, such as withdrawal symptoms, inefficiency, and avoidance, may differentially contribute to academic burnout, warranting further investigation in future research. The influence of mobile phone addiction on academic burnout may be attributed to two main factors. First, according to Cognitive Load Theory, cognitive activities during learning consume limited cognitive resources. Excessive use of mobile phones distracts attention, reducing the ability of Chinese college students to focus on learning tasks⁵⁹. This may be particularly pronounced in high-pressure academic environments and in collectivist cultural contexts, where multitasking and social

comparison are more common. In Chinese universities, strong collectivist norms and achievement-oriented competition may intensify students' reliance on the digital interaction and magnify the psychological cost of distraction, reinforcing the cultural relevance of these findings. In addition, mobile phone addiction may indirectly lead to academic burnout by affecting the sleep quality of Chinese college students^{31,32}. Prolonged nighttime use inhibits melatonin secretion, disrupts circadian rhythms, and reduces sleep quality, which in turn diminishes daytime attention, motivation, and learning efficiency, ultimately increasing the risk of academic burnout. These mechanisms highlight both cognitive and physiological pathways linking mobile phone addiction to academic burnout^{60,61}.

The Mediating Role of learning engagement

This study found that learning engagement plays a significant mediating role between mobile phone addiction and academic burnout, supporting Hypothesis 2. This mediation effect underscores the motivational mechanism specified by self-determination theory: when autonomy, competence, and relatedness needs are frustrated by excessive phone use, intrinsic motivation and engagement decline, leading to burnout. Specifically, mobile phone addiction can significantly reduce Chinese college students' learning engagement, which in turn increases academic burnout¹¹. The underlying reasons may involve the following aspects. First, according to the Self-Determination

Theory, when individuals' autonomy needs, competence needs, and relationship needs are not met, their intrinsic motivation is weakened⁶². Excessive use of mobile phones can lead Chinese college students to indulge in virtual social interaction and entertainment, reduce their effective interaction with others in real life and leave their relatedness needs unmet; second, frequent use of mobile phones can distract attention, reduce learning efficiency, and make it difficult for Chinese college students to experience a sense of accomplishment in learning, thus weakening the competency needs. These factors together lead to a decrease in learning engagement, which in turn increases the likelihood of academic burnout⁵².

Finally, as a positive learning state, learning engagement can enhance Chinese college students' self-efficacy and enhance their learning motivation⁶³. When Chinese college students devote themselves to learning, they can better understand and master knowledge, improve their academic performance, and then enhance their self-efficacy, further stimulate their learning motivation, and reduce their academic burnout¹¹.

The Moderating Role of physical exercise

The results of this study show that physical exercise plays a significant moderating role between mobile phone addiction and learning engagement, which is consistent with Hypothesis 3. Specifically, in the high physical exercise group, the negative impact of mobile phone addiction on learning engagement was not

significant, whereas it was substantial in the low physical exercise group. This effect can be interpreted through stress-and-coping and emotion regulation perspectives: Regular physical exercise enhances emotion regulation and self-control, replenishes depleted psychological resources, and improves executive functioning, thereby buffering the negative cognitive and emotional consequences of excessive mobile-phone use. These mechanisms align with evidence that physical activity strengthens self-discipline and resilience, mitigating burnout risk⁶⁴. From a physiological mechanism, physical exercise can promote the brain to secrete neurotransmitters such as endorphins and dopamine, which can not only improve emotional state, but also improve attention and cognitive ability⁴⁵. Therefore, Chinese college students who often take part in physical exercise can better control their attention and reduce the interference of mobile phones on learning when facing mobile phone addiction, to maintain a high level of learning engagement. In addition, physical exercise can also enhance Chinese college students' self-control ability and self-discipline⁶⁵. By formulating and implementing physical exercise plans, Chinese college students can cultivate their willpower and self-discipline habits, and these abilities and habits can also be transferred to study to help them better manage their study time and behavior and improve their study engagement¹⁶. Furthermore, the moderating role of physical exercise identified in this study offers meaningful implications for digital wellness development and

academic burnout prevention in higher education. These findings support the value of incorporating regular exercise into university well-being initiatives, digital-use management strategies, and student support systems to promote healthier technology habits and academic sustainability.

Conclusions and Implications

Conclusions

Based on data from 677 Chinese college students across three universities in Shandong Province, this study constructed and validated a moderated mediation model, elucidating the mechanism by which mobile phone addiction influences academic burnout among Chinese college students. By integrating the I-PACE and COR frameworks, this model provides a comprehensive explanation of how cognitive load, motivational loss, and resource depletion jointly lead to burnout, while physical exercise serves as a protective moderator restoring self-regulatory capacity. The results indicate a significant positive correlation between mobile phone addiction and academic burnout, such that higher levels of mobile phone addiction are associated with increased academic burnout. Learning engagement was found to mediate this relationship significantly, with mobile phone addiction contributing to academic burnout through reduced learning engagement. Furthermore, physical exercise moderates the relationship between mobile phone addiction and learning engagement, with higher levels of physical exercise mitigating the negative impact of mobile phone addiction

on learning engagement. Specifically, in the high physical exercise group, the negative impact of mobile phone addiction on learning engagement was not significant, whereas in the low physical exercise group, mobile phone addiction had a substantial negative effect on learning engagement.

Implications

To effectively reduce academic burnout among Chinese college students and promote their mental, physical, and academic development, multi-level, theory-based interventions are recommended. At the institutional level, universities should implement digital well-being education and self-regulation education programs grounded in cognitive-behavioral evidence, establish clear norms for mobile phone use in learning spaces, and regulate non-academic entertainment and social-media access during instructional hours. Empirical research shows that structured digital-literacy curricula can significantly reduce students' problematic phone use and improve self-control⁶⁶. Rather than relying solely on behavioral restriction, universities should enhance management of campus networks and mobile-phone use. Specifically, universities may guide students to balance learning and leisure through self-monitoring tools and digital-use feedback systems, supported by campus counseling centers. Additionally, universities should strengthen mental health education and counseling services by improving the mental health curriculum, incorporating modules on digital-well-being and academic-stress

management, and providing evidence-based psychological counseling.

At the teacher level, educators should innovate teaching methods and strengthen feedback and motivation strategies to sustain students' learning engagement. Teacher interventions supported by achievement-motivation and self-determination research can foster intrinsic engagement and reduce burnout⁶². Leveraging modern information technology and designing challenging, engaging tasks tailored to different disciplines can enhance participation, sense of accomplishment, and learning outcomes. Empirical classroom studies show that active-learning designs significantly increase vigor and dedication—the core dimensions of engagement¹.

At the student level, students should be encouraged to participate in regular physical exercise through structured PE classes and extracurricular sports clubs. Evidence from large-sample studies indicates that regular moderate-to-vigorous exercise enhances emotional regulation and resilience, thereby lowering burnout risk⁶⁷⁻⁶⁹. Integrating digital-self-control and mindfulness-based training with physical exercise, as suggested by self-regulated-learning theory and supported by experimental studies, can help students manage study time, improve engagement, and reduce burnout. Therefore, universities should enforce physical-fitness assessments while ensuring that physical exercise policies emphasize health promotion and psychological

well-being rather than punishment, motivating students to maintain lifelong exercise habits. Overall, implementing these empirically supported, multi-level interventions, spanning institutional policies, teaching practices, and student self-management, can provide a comprehensive framework for mitigating mobile-phone addiction and academic burnout in higher-education settings.

Limitations

This study has several limitations that should be noted. First, the reliance on self-report measures may introduce social desirability bias, potentially leading participants to underreport their mobile phone addiction and academic burnout or overreport their learning engagement, which could affect the reliability and validity of the findings. Future research should consider employing multiple data collection methods, such as behavioral tracking or observational studies, to enhance the robustness of the results. Second, although the sample size is adequate, the participants were recruited from a single province, which may limit the generalizability of the findings. Future research should consider multi-region or stratified random sampling to enhance representativeness and examine potential regional or institutional variations in the proposed model. Third, the cross-sectional design of this study precludes definitive conclusions about causality and does not allow for examination of temporal changes in the variables. Future research should employ longitudinal tracking or experimental intervention designs to examine causal processes and

dynamic trajectories, thereby strengthening the explanatory power of the proposed model. Additionally, future research could explore the differential impacts of various types of physical exercise (e.g., intensity, frequency, duration) on academic burnout to further elucidate the underlying mechanisms.

Data availability: Data are provided within the manuscript or supplementary information files.

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